SIMATS ENGINEERING
ASSESSMENT TOOL 3

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COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and indepth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

1. Star Topology vs Bus Topology

Criteria	Star Topology	Bus Topology
Structure	All devices connect individually to a central switch/hub.	All devices share a single backbone cable.
Cost	Higher because it requires a central switch/hub and more cables.	Lower because only one main backbone cable is required.
Installation	Relatively easy but requires more cabling.	Simple and easy to install.
Reliability	High. Failure of one node or cable normally affects only that node.	Lower. Failure of the backbone can affect the entire network.
Performance	Better because the central switch manages communication and reduces collisions.	Performance decreases as more devices share the common medium.
Maintenance	Easy to troubleshoot because individual connections can be isolated.	Difficult when a backbone fault occurs.
Scalability	Good, but limited by switch ports.	Limited because adding devices increases traffic on the shared cable.
Best Use	Modern LANs, offices, laboratories and enterprise networks.	Small or simple networks where low cost is important.

Analysis:

Star topology is more expensive than bus topology because every device requires an individual connection to the central switch. However, the additional cost provides better reliability and performance. If one workstation or its cable fails, the remaining devices can continue communicating. In a bus topology, all devices use the same backbone cable. Therefore, a backbone failure can bring down the entire network. Also, increasing the number of devices increases shared traffic and can reduce performance.

Example:

A university computer laboratory is better suited to a star topology because each computer can be connected to a central switch. If one computer fails, the other computers remain operational.

Conclusion:

Star topology is preferred for modern LANs because its higher installation cost is justified by better reliability, performance, troubleshooting, and manageability.

2. Mesh Topology vs Star Topology – Fault Tolerance

Criteria	Mesh Topology	Star Topology
Connectivity	Nodes have multiple direct connections.	Nodes connect through a central switch/hub.

Fault Tolerance	Very high. Multiple alternate paths are available.	Moderate. Individual node failure does not affect others.
Link Failure	Traffic can use another available path.	A failed individual link affects that connected device only.
Central Device Dependency	No single central device is required.	Central switch/hub is critical.
Reliability	Extremely high.	High for individual-link failures.
Cost	Very high due to many links and ports.	Lower than full mesh.
Complexity	High.	Relatively simple.
Criteria	**Mesh Topology**	**Star Topology**
Maintenance	Difficult because of many connections.	Easier to identify and isolate faults.
Suitable Environment	Critical networks requiring redundancy.	Offices, labs and normal LANs.

Analysis

Mesh topology provides very high fault tolerance because each node can have multiple connections. When one link fails, communication can continue through an alternate path. This makes mesh topology suitable for critical environments where network availability is extremely important.

Star topology provides moderate fault tolerance. Failure of an individual node or cable normally does not affect other devices. However, failure of the central switch can stop communication for all connected devices.

Example

A critical backbone network may use mesh connections because alternate paths are necessary. A computer laboratory can use star topology because it provides a good balance between reliability and cost.

Conclusion

Mesh provides greater fault tolerance, while star provides a better balance between reliability, cost, and manageability.

3. Bus Topology vs Mesh Topology – Installation Complexity

Criteria	Bus Topology	Mesh Topology
Cable Requirement	One main backbone cable.	Multiple connections between nodes.
Installation	Simple.	Complex.
Number of Connections	Low.	Very high.
Cost	Low.	High.
Configuration	Easy to configure.	Requires careful configuration.
Scalability	Limited.	Complex as nodes increase.
Maintenance	Backbone faults can be difficult to locate.	Many individual connections must be maintained.

Best Application	Small/simple networks.	Critical networks requiring redundancy.

Analysis

Bus topology is simple and economical because all devices share a single backbone cable. Therefore, fewer cables and connections are required.

Mesh topology is considerably more complex because devices require multiple links. As the number of devices increases, the number of connections increases significantly. This makes installation timeconsuming and expensive.

Example

For a small temporary laboratory network, a simple bus arrangement can reduce installation cost. For a critical communication backbone, mesh is more suitable because reliability is more important than installation simplicity.

Conclusion

Bus topology is easier and cheaper to install, whereas mesh topology provides higher redundancy but requires significantly greater installation effort and cost.

4. Hybrid Topology vs Star Topology – Scalability

Criteria	Hybrid Topology	Star Topology
Structure	Combines two or more topologies such as star, mesh and bus.	Devices connect to a central switch/hub.
Scalability	Very high. Different sections can be expanded independently.	Good but dependent on central-device capacity.
Flexibility	Very high. Different areas can use different topologies.	Moderate. Same basic structure is maintained.
Expansion	New network segments can be added without major changes to existing sections.	Additional hardware may be needed when switch ports are exhausted.
Reliability	Can provide high reliability by combining redundant topologies.	Good, but central-device failure is a major concern.
Cost	Depends on the technologies combined.	Generally moderate.
Complexity	Higher because multiple topology types are managed.	Lower.
Best Application	Large campuses, enterprises and complex networks.	Offices, classrooms and LANs.

Analysis

Hybrid topology is highly scalable because it combines different topology structures according to the requirements of different network sections. New segments can be added without redesigning the entire existing network.

Star topology is also scalable, but expansion depends on the available ports of the central switch. When all ports are occupied, another switch or additional hardware must be introduced.

Example

A large university campus can use a hybrid topology:

Campus Backbone → Mesh/Ring → Building Distribution → Star → Laboratory Bus/Tree This allows each section to use the topology that best matches its requirements.

Conclusion

Hybrid topology provides greater scalability and flexibility than star topology, making it highly suitable for large organizations and campus networks.

5. Maintenance Comparison of Star, Mesh, Bus and Hybrid Topologies

Criteria	Star	Mesh	Bus	Hybrid
Maintenance Difficulty	Low	High	Medium/High	Medium
Fault Identification	Easy	Complex	Difficult for backbone faults	Moderate
Troubleshooting	Easy	Timeconsuming	Backbone troubleshooting can be difficult	Requires skilled administration
Reliability	High	Very high	Lower	High depending on design
Network Management	Simple	Complex	Simple structure but difficult fault isolation	More complex
Scalability	Good	Complex and expensive	Limited	Excellent
Best Use	LANs and offices	Critical networks	Small networks	Large enterprise/campus networks

Analysis

Star topology is relatively easy to maintain because each device has an individual connection to the central switch. If a computer or cable fails, the administrator can isolate that specific connection without affecting the rest of the network.

Mesh topology is highly reliable but difficult to maintain because it contains many connections. Troubleshooting becomes time-consuming as the number of devices increases.

Bus topology is simple in structure, but a fault in the main backbone cable can affect the entire network and can be difficult to locate.

Hybrid topology has moderate maintenance complexity. It provides flexibility and scalability, but administrators must understand and monitor multiple topology structures.

Example

For a university campus, hybrid topology is suitable because different areas have different requirements. The backbone can use redundant mesh/ring connections, while individual buildings can use star topology.

Overall Comparative Analysis

Topology	Cost	Reliability	Fault Tolerance	Installation	Scalability	Maintenance
Star	Medium	High	Medium	Easy	Good	Easy
Bus	Low	Low/Medium	Low	Very Easy	Limited	Moderate

Mesh	Very High	Very High	Very High	Difficult	Complex	Difficult
Hybrid	Medium/High	High	High	Moderate/Complex	**Excellent**	Moderate